WORKSHEET 3.7 : CARTESIAN EQUATION OF A PLANE

1a) $7x + y - z + D = 0$

$7(2) + (1) - (-3) + D = 0$

$18 + D = 0$

$D = -18$

$\therefore \boxed{7x + y - z - 18 = 0}$

b) $x + D = 0$

$(5) + D = 0$

$D = -5$

$\therefore \boxed{x - 5 = 0}$

c) $2x + 3z + D = 0$

$2(0) + 3(-2) + D = 0$

$D = 6.$

$\therefore \boxed{2x + 3z + 6 = 0}$

d) $2x - y + 4z + D = 0$

$2(\phi) - (\phi) + 4(\phi) + D = 0$

$D = 0$

$\therefore \boxed{2x - y + 4z = 0}$

2a) $\vec{n}$ // y-axis. $\therefore \vec{n} = (0, 1, 0)$

$y + D = 0$

$-2 + D = 0$

$D = 2$

$\therefore \boxed{y + 2 = 0}$

b) $\perp$ to xy-plane

$\therefore$ // to z-axis.

$\therefore \vec{n} = (0, 0, 1)$

$z + D = 0.$

$(3) + D = 0$

$D = -3$

$\therefore \boxed{z - 3 = 0}$

c) // to normal of $x - y - 2z + 19 = 0$

$\therefore \vec{n} = (1, -1, -2)$

$1x - y - 2z + D = 0$

$1(1) - (-2) - 2(3) + D = 0$

$1 + 2 - 6 + D = 0$

$D = 3$

$\therefore \boxed{x - y - 2z + 3 = 0}$

3a) $\vec{n} = (A, B, C)$ $\therefore Ax + By + Cz + D = 0$

pt $(0,0,0)$ $\therefore D = 0$.

$\therefore \boxed{Ax + By + Cz = 0}$

b) If $D = 0$, then passes through the origin.

5a) $\vec{r} = (-8, -1, 8) + s(-5, 1, 4) + t(3, 2, -4)$

$\vec{n} = (-5, 1, 4) \times (3, 2, -4)$

$= (-12, -8, -13)$

OR $= (12, 8, 13)$

$-5 \quad 1 \quad 4 \quad -5 \quad 1 \quad 4$

$3 \quad 2 \quad -4 \quad 3 \quad 2 \quad -4$

$= (-12, -8, -13)$

$\therefore 12x + 8y + 13z + D = 0$

$12(-8) + 8(-1) + 13(8) + D = 0$

$D = 0$.

$\therefore \boxed{12x + 8y + 13z = 0}$

b) $\vec{r} = (-2, -2, 5) + s(3, 1, -1) + t(4, 1, -4)$

$\vec{n} = (3, 1, -1) \times (4, 1, -4)$

$= (-3, 8, -1)$

OR $= (3, -8, 1)$

$3 \quad 1 \quad -1 \quad 3 \quad 1$

$4 \quad 1 \quad -4 \quad 4 \quad 1$

$= (-3, 8, -1)$

$3x - 8y + z + D = 0$

$3(-2) - 8(-2) + (5) + D = 0$

$D = -15$

$\therefore \boxed{3x - 8y + z - 15 = 0}$

5c) $\vec{r} = (2,0,0) + s(0,4,0) + t(0,0,-3)$

$4\hat{j}$ $\quad -3\hat{k}$

$\therefore \vec{n} = (1,0,0)$ ← a vector ⊥ both $\hat{j}$ & $\hat{k}$.

$x + D = 0$
$2 + D = 0$
$D = -2$

$\therefore \boxed{x - 2 = 0}$

d) $\vec{r} = (-8,2,0) + s(4,0,3) + t(0,-2,-5)$

$\vec{n} = (4,0,3) \times (0,-2,-5)$
$= (6,20,-8)$
OR $= (3,10,-4)$

$4 \quad 0 \quad 3 \quad 4 \quad 0$
$0 \quad -2 \quad -5 \quad 0 \quad -2$
$= (6,20,-8)$

$3x + 10y - 4z + D = 0$
$3(-8) + 10(2) + D = 0$
$D = 4$

$\therefore \boxed{3x + 10y - 4z + 4 = 0}$

e) $\vec{r} = (2,-11,-17) + s(0,5,13) + t(0,3,10)$

yz-plane yz-plane ⊥ to yz-plane = x-axis!

$\vec{n} = (0,5,13) \times (0,3,10)$
$= (11,0,0)$
OR $= (1,0,0)$ = x-axis ✓

$0 \quad 5 \quad 13 \quad 0 \quad 5$
$0 \quad 3 \quad 10 \quad 0 \quad 3$
$= (11,0,0)$

$\therefore x + D = 0$
$2 + D = 0$
$D = -2$

$\boxed{x - 2 = 0}$

f) $\vec{r} = (13,0,-12) + s(-1,8,-4) + t(11,3,-12)$

$\vec{n} = (-1,8,-4) \times (11,3,-12)$
$= (-84,-56,-91)$
OR $(12,8,13)$

$-1 \quad 8 \quad -4 \quad -1 \quad 8$
$11 \quad 3 \quad -12 \quad 11 \quad 3$
$= (-84,-56,-91)$

$12x + 8y + 13z + D = 0$
$12(13) + 8(0) + 13(-12) + D = 0$
$D = 0$

$\therefore \boxed{12x + 8y + 13z = 0}$

6a) Pt (4, 2, 1)

$\vec{n} = (3, 4, -2) \times (-2, 4, -3)$

$= (-4, 13, 20)$

OR $= (4, -13, -20)$

$4x - 13y - 20z + D = 0$

$4(4) - 13(2) - 20(1) + D = 0$

$D = 30$

$3/4\ \ -2\ \ 3\ \ 4\ \ -2/$
$-2\ \ 4\ \ -3\ \ -2\ \ 4\ \ -3$

$= (-4, 13, 20)$

$\therefore$ **$4x - 13y - 20z + 30 = 0$**

b) Pt (0, 2, 5)

$\vec{n} = (0, -1, 3) \times (-2, -3, 0)$

$= (9, -6, -2)$

$9x - 6y - 2z + D = 0$

$9(0) - 6(2) - 2(5) + D = 0$

$D = 22$

$9/-1\ \ 3\ \ 0\ \ -1\ \ 3/$
$-2\ \ -3\ \ 0\ \ -2\ \ -3\ \ 0$

$= (9, -6, -2)$

$\therefore$ **$9x - 6y - 2z + 22 = 0$**

7a) (1, 1, −1)
(1, 2, 3)
(3, −1, 2)

$\vec{a} = (0, 1, 4)$

$\vec{b} = (2, -3, -1)$

$\vec{n} = (0, 1, 4) \times (2, -3, -1)$

$= (11, 8, -2)$

$11x + 8y - 2z + D = 0$

$11(1) + 8(1) - 2(-1) + D = 0$

$D = -21$

$9/1\ \ 4\ \ 0\ \ 1\ \ 4/$
$2\ \ -3\ \ -1\ \ 2\ \ -3\ \ 1$

$= (11, 8, -2)$

$\therefore$ **$11x + 8y - 2z - 21 = 0$**

b) (2, −2, 4)
(1, 1, −4)
(3, 1, −6)

$\vec{a} = (1, -3, 8)$

$\vec{b} = (2, 0, -2)$

$\vec{n} = (1, -3, 8) \times (2, 0, -2)$

$= (6, 18, 6)$

OR $= (1, 3, 1)$

$x + 3y + z + D = 0$

$(2) + 3(-2) + (4) + D = 0$

$D = 0$

$1/-3\ \ 8\ \ 1\ \ -3\ \ 8/$
$2\ \ 0\ \ -2\ \ 2\ \ 0\ \ -2$

$= (6, 18, 6)$

$\therefore$ **$x + 3y + z = 0$**

7c) $(1,1,1)$
$(-1,1,1)$
$(2,1,2)$

$\vec{a} = (2,0,0)$
$\vec{b} = (1,0,1)$
$\vec{n} = (2,0,0) \times (1,0,1)$
$= (0,-2,0)$
or $(0,1,0)$

$\therefore y + D = 0$
$(1) + D = 0$
$D = -1$

$\therefore \boxed{y - 1 = 0}$

d) $(1,3,0)$
$(0,5,2)$
$(3,4,-2)$

$\vec{a} = (1,-2,-2)$
$\vec{b} = (3,-1,-4)$
$\vec{n} = (1,-2,-2) \times (3,-1,-4)$
$= (6,-2,5)$

$\therefore 6x - 2y + 5z + D = 0$
$6(1) - 2(3) + 5(0) + D = 0$
$D = 0$

$\therefore \boxed{6x - 2y + 5z = 0}$

8) x-axis

$P + (4,-2,1)$

another point on x-axis
$(0,0,0)$

$\therefore \vec{a} = (1,0,0)$
$\vec{b} = (4,-2,1)$

$\vec{n} = (1,0,0) \times (4,-2,1)$
$= (0,-1,-2)$
OR $(0,1,2)$

$y + 2z + D = 0$
$(0) + 2(0) + D = 0$
$D = 0$

$\therefore \boxed{y + 2z = 0}$

9) $P + (2,0,-3)$

$\vec{a} = (1,2,3)$
$\vec{b} = (-3,4,2)$

$\vec{n} = \vec{a} \times \vec{b}$
$= (-8,-11,10)$
OR
$= (8,11,-10)$

$\begin{matrix} 1 & 2 & 3 & 1 & 2 \\ -3 & 4 & 2 & -3 & 4 \end{matrix}$
$= (-8,-11,10)$

$8x + 11y - 10z + D = 0$
$8(2) + 11(0) - 10(-3) + D = 0$
$D = -46$

$\therefore \boxed{8x + 11y - 10z - 46 = 0}$

10a) $\vec{n_1} = (1, 3, -1)$ $\vec{n_2} = (2, 6, -2)$ $\therefore \vec{n_2} = 2\vec{n_1}$ ✓

since cartesian equations are not scalar multiples.

i.e. $x + 3y - z - 2 = 0$. $\times 2 = 2x + 6y - 2z - 4 = 0$

↳ not -8.

∴ not coincident but parallel

b) $\vec{n_1} = (2, 1, 1)$ $\vec{n_2} = (6, 2, 2)$

Let $\vec{n_1} = k\vec{n_2}$

$2 = 6k$ $1 = 2k$ $1 = 2k$

$\frac{1}{3} = k$ $\frac{1}{2} = k$ $\frac{1}{2} = k$

since k is not consistent, not parallel ∴ neither

c) $3x - 3y + z - 2 = 0$

$6x - 6y + 2z - 4 = 0$ ↵ $\times 2$

∴ scalar multiples. ∴ coincident planes

d) $2x - 4y + 2z - 6 = 0$

$3x - 6y + 3z - 9 = 0$ ↵ $\times \frac{3}{2}$

∴ since equations are scalar multiples ∴ coincident planes

11a) $2x - y + 3z - 24 = 0$

find 3 points. (easy: x-int, y-int, z-int)

x-int $2x - 24 = 0$, $x = 12$, $(12, 0, 0)$

y-int $-y - 24 = 0$, $y = -24$, $(0, -24, 0)$

z-int $3z - 24 = 0$, $z = 8$, $(0, 0, 8)$

$\vec{a} = (12, 24, 0)$ $\vec{b} = (12, 0, 8)$

$\therefore \vec{r} = (12, 0, 0) + s(12, 24, 0) + t(12, 0, 8),\ s, t \in \mathbb{R}$

11b) $3x - 5z + 15 = 0$.

Find 3 points:

x-int // $3x + 15 = 0$, $x = -5$, $(-5, 0, 0)$

z-int // $-5z + 15 = 0$, $z = 3$, $(0, 0, 3)$

y can be anything. Let $y = 1$, $(0, 1, 3)$

$\vec{a} = (5, 1, 3)$ $\quad \vec{b} = (0, 1, 0)$

$\therefore \vec{r} = (-5, 0, 0) + s(5, 1, 3) + t(0, 1, 0)$, $s, t \in \mathbb{R}$

12a) if $\not\parallel$, then $\vec{n} \cdot \vec{m} = 0$.

$\vec{n} = (4, 1, -1)$ $\quad \vec{m} = (1, -2, 2)$

$\vec{n} \cdot \vec{m} = 4 - 2 - 2$

$= \emptyset$ ✓

$\therefore$ line is $\not\parallel$ to plane.

Check point:

LS $= 4x + y - z - 10$ $\quad$ RS $= \emptyset$

$= 4(3) + (\emptyset) - (2) - 10$

$= \emptyset$

$\therefore$ LS = RS

$\therefore$ line lies on the plane

b) $\vec{n} = (4, 1, -1)$ $\quad \vec{m} = (-3, 2, -10)$

$\vec{n} \cdot \vec{m} = -12 + 2 + 10$

$= \emptyset$ ✓

$\therefore$ line is $\not\parallel$ to plane

check pt $(0, -5, 0)$

LS $= 4x + y - z - 10$ $\quad$ RS $= \emptyset$.

$= 4(\emptyset) - 5 - (\emptyset) - 10$

$= -15$

LS $\neq$ RS

$\therefore$ line does not lie in the plane.

c) $\vec{n} = (4, 1, -1)$ $\quad \vec{m} = (4, -1, 1)$

$\vec{n} \cdot \vec{m} = 16 - 1 - 1$

$= 14 \neq 0$

$\therefore$ line not $\not\parallel$ to plane

$\therefore$ line does not lie in the plane

13 a) $\vec{n_1} = (2, 3, -1)$

$\vec{n_2} = (1, 2, \phi)$

$$\cos\theta = \frac{\vec{n_1} \cdot \vec{n_2}}{|\vec{n_1}||\vec{n_2}|}$$

$$\cos\theta = \frac{2 + 6 + \phi}{\sqrt{14}\sqrt{5}}$$

$$\cos\theta = \frac{8}{\sqrt{70}}$$

$$\boxed{\theta = 17^\circ}$$

b) $\vec{n_1} = (1, -1, -1)$

$\vec{n_2} = (2, 3, -1)$

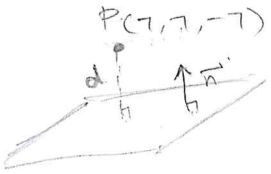
$$\cos\theta = \frac{\vec{n_1} \cdot \vec{n_2}}{|\vec{n_1}||\vec{n_2}|}$$

$$\cos\theta = \frac{2 - 3 + 1}{\sqrt{3}\sqrt{14}}$$

$$\cos\theta = \phi$$

$$\boxed{\theta = 90^\circ}$$

11 a)



Need a point P_0 on the plane.

$6y - z + 5 = 0$ $\qquad \vec{n} = (0, 6, -1)$

$(0, 0, 5)$

$$\therefore \vec{P_0P} = (7, 7, -7) - (0, 0, 5)$$

$$= (7, 7, -12)$$

$$d = \frac{|\vec{P_0P} \cdot \vec{n}|}{|\vec{n}|}$$

$$= \frac{|(7, 7, -12) \cdot (0, 6, -1)|}{\sqrt{37}}$$

$$= \frac{|42 + 12|}{\sqrt{37}}$$

$$\boxed{d = \frac{54\sqrt{37}}{37}}$$

b) $P_0(0, 0, 10)$

$P(3, 2, 1)$ $\Big\}$ $\vec{P_0P} = (3, 2, -9)$

$\vec{n} = (3, 2, 1)$

$$d = \frac{|\vec{P_0P} \cdot \vec{n}|}{|\vec{n}|}$$

$$= \frac{|9 + 4 - 9|}{\sqrt{14}}$$

$$\boxed{d = \frac{4\sqrt{14}}{14} = \frac{2\sqrt{14}}{7}}$$